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Electronic Practical Work

Microcontrollers Security

RDP Protection

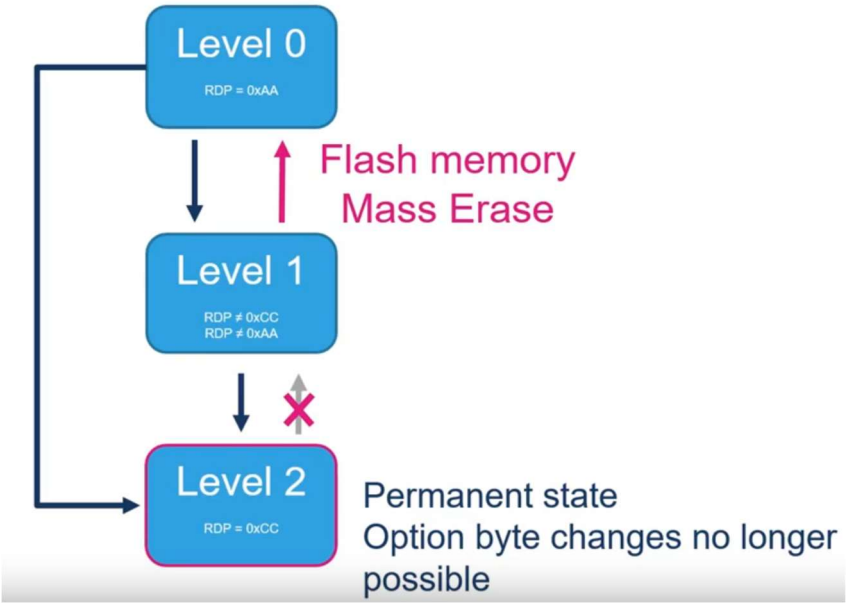
Jean-Jacques Meneu

RDP stands for Read Out Protection. This is a protection of the Flash memory to prevent copying of the code, reverse engineering, inserting code, using development tools to access the program…
RDP has three levels on most microcontrollers (the most secure ones with a 'trust zone' have 4 levels, which will be seen later)

- ✓ RDP level 0 : no protection and is the default level
  - Reading, writing and erasing Flash, SRAM
  - Option bytes can be modified
- ✓ RDP level 1 :
  - If the boot is the main Flash (so neither SRAM nor bootloader) and no debugger access is detected
    - Reading, writing and erasing of Flash and SRAM are possible internally (in other words, the code can perform these actions, but not from the outside)
    - Changing option bytes is possible (via the bootloader)
  - If booting is done via the bootloader or if debugger access is detected
    - Reading, writing and erasing of Flash and SRAM (not in all cases for SRAM) are blocked and a HardFault interrupt is generated.
    - Changing option bytes is possible
- ✓ RDP level 2 :

  Flash is transformed into ROM, it is impossible to read, program and communicate with the microcontroller. Booting can only be done from Flash memory, option bytes cannot be modified and RDP regression is impossible (in other words, no RDP level 2 to level 1 or 0 is impossible)

The figure below summarizes the possible changes between the different protection levels. Changing from level 1 to level 0 allows you to erase all the contents of the FLASH.



NEVER SET THE MICROCONTROLLER TO LEVEL 2 DURING THE TP !!!!!!!!!!! OTHERWISE YOU MUST BUY A NEW CARD

Jean-Jacques Meneu

Protections may vary depending on the microcontroller, so it is necessary to read the specifications to know exactly the protection mechanisms depending on the component used.

Notes :

- ✓ Enabling an RDP level often requires a power cycle (in other words, going from 0V to 3V). Check in the reference manual
- ✓ With the RDP level at 1, it is possible to read the device registers and in some cases the contents of the SRAM which is problematic for security. In this case, the developer can be forced to execute the code only in the internal registers without using the SRAM.

Work to be provided

The goal of the manipulation is simply to see the impact of a regression of the RDP from 1 to 0.

- ✓ Write the program that makes the LED flash every second.

```
*((uint32_t *) 0x20000020) = 0xDEADBEEF;

while (1)
{
    HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
    HAL_Delay(1000);
    cpt++;
```

- ✓ Run the code and locate the address of the cpt variable that was declared as global
- ✓ Using the STM32CubeProgrammer, set the RDP to level 1.
- ✓ Check the contents of the Flash and SRAM after performing a power cycle if necessary. What is the difference between the contents in the Flash memory and the SRAM? Check if the program is debuggable.
- ✓ Regress the RDP from level 1 to level 0. What value is stored in the Flash?